



Encodings of multimodal conversational behavior predict trait impressions

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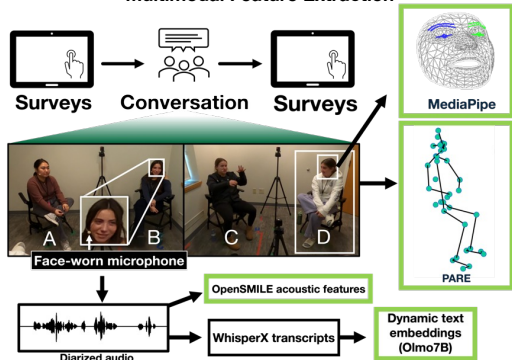
Introduction

People infer others' personality traits from dynamic social signals spread across many expressive channels. What aspects of personality are distributed across these channels (i.e., facial expressivity, body expressivity, voice quality and dynamics, semantic content)?

Methods

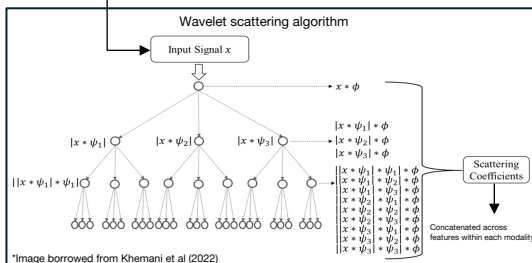
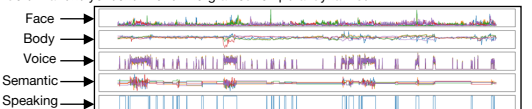
- Group conversations dataset:** 40 groups; 2-4 person; N=120; 30 minutes each; unprompted/unconstrained
- Pre- and post-conversation Big-5 Personality Index (BFII)** for self and other
- Multimodal feature extraction:** Dynamic facial expression, 3D body pose, vocal acoustics, and dynamic semantic embeddings from transcribed speech; resampled to 60Hz

Multimodal Feature Extraction



Multivariate time series encoding with wavelet scattering

The wavelet scattering transform is a cascading frequency decomposition that repeatedly convolves an input time series with a Morlet wavelet filter bank, producing 0th-, 1st-, and 2nd-order coefficients that capture signal structure across a hierarchy of timescales. The result is a time-collapsed representation that is stable to small deformations yet sensitive to fine-grained temporal dynamics.



For each feature of each time series, we applied the wavelet scattering algorithm to derive concatenated vectors of scattering coefficients, encoding the "texture" of each modality's dynamics. The coefficients were then re-weighted to maximize reliable between-person variation.

Inter-subject Representational Similarity Analysis

Use a series of linear mixed effects models, we treat subject's personality similarity across each BFI dimension (OCEAN), locus (self & other) and timepoint (pre- & post-conversation) as a function of cosine similarity between each pairing's modality-specific scattering vectors. We included scattering vectors derived from binary speaking status time series as a covariate and controlled for random effects of subject and group IDs.

Model formula

$$\text{Personality Sim.} \sim \beta_0 + \beta(\text{Face}) + \beta(\text{Body}) + \beta(\text{Voice}) + \beta(\text{Semantic}) + \beta(\text{Speaking}) + \beta(\text{Same Group}) + \text{Subject \& Group Random Intercepts}$$

Results

Beta-values for modality predictors: * = < .001; ** = < .01; *** = < .001

O = Openness; C = Conscientiousness; E = Extraversion; A = Agreeableness; N = Neuroticism

Rating	Face	Body	Voice	Semantic	Speaking
Pre-Self	O -.003	.006	.004	-.008	.009
	Self C -.003	-.004	-.030***	.008	.045***
	E .060***	.065***	.022*	.039***	.016
	A .003	-.006	-.007	-.010	.006
	N -.005	-.010	.003	.001	.021**
Post-Self	O .012	.003	-.014	-.005	.015
	Self C .006	.041***	.050***	.005	.020*
	E .070***	.042***	.034***	-.001	.010
	A -.006	-.003	-.002	-.013	.011
	N .017*	-.005	.000	-.006	-.004
Pre-Other	O .033***	.023***	-.014	.006	-.005
	Other C -.002	.002	-.003	.000	.007
	E .076***	.011	-.042***	.039***	.064***
	A -.002	.012	-.006	-.002	.013
	N -.009	-.003	-.007	-.005	.020*
Post-Other	O .069***	.003	.011	.023***	.024***
	Other C .035***	.064***	.019*	.021***	.010
	E .074***	.053***	.012	.045***	.053***
	A .025***	-.005	-.006	-.006	-.005
	N -.001	-.013	.005	-.007	.000

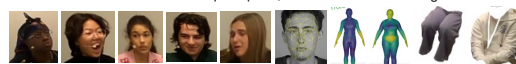
- Expressive similarity predicts personality similarity
 - effects vary by modality, trait, rater, timing
- Extraversion most robustly predicted
- Post-conversation, other-ratings show strongest effects; self-ratings less predictive

Conclusions

Variable modality contributions predicting BFI dimensions suggest that different traits may be more perceptible through different channels of social signaling. Trait impressions may be partially tacit and interaction-dependent, rather than being derived from biographical knowledge or static features alone.

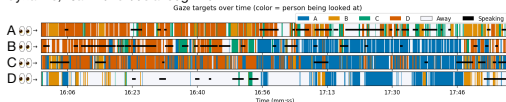
Future Directions: Static social features

Static features (facial structure, clothing, body composition, skin tone) likely make dissociable contributions to social perception; extract and include as regressors.



Future Directions: Social information dynamics

Future and ongoing work will use deep learning and information-theoretic tools to analyze how social information is distributed across time, people, and modalities during social interaction. Studying information dynamics of embodied interaction is a precursor to more a comprehensive understanding of the neural computations that support dynamic, real-world social cognition.



References

- Andrew, M., Angles, T., Enoch, G., et al. (2018). *Scattering Transform in Python*. *arXiv:1812.11214*.
- Ban, M., Hui, J., Han, T., & Zisserman, A. (2020). *WhisperX: Time-Accurate Speech Transcription of Long-Form Audio*. *INTERSpeech*.
- Epstein, F., Wimmer, M., & Zisserman, A. (2019). *OpenSMILE: The Munich Versatile and Fast Open-Source Audio Feature Extractor*. *ACM Multimedia*, 145B-1462.
- Groenewold, D., Bittig, L., Wain, P., et al. (2020). *CLIM: Accelerating the Science of Language Models*. *arXiv:2002.00808*.
- Kozinske, M., Huang, C.-H. P., Hilliges, O., & Black, M. J. (2021). *PAPE: Per Attention Programmer for 3D Human Body Estimation*. *arXiv:2104.06527*.
- Li, S., Tang, J., Nath, H., et al. (2019). *MediaPipe: A Framework for Building Perception Pipelines*. *arXiv:1906.08172*.
- Mallat, S. (2011). *Group Invariant Scattering*. *arXiv:1107.2088*.
- Khemani, V., Asari, M. H., & Pecht, M. G. (2020). *Learnable wavelet scattering networks: Applications to fault diagnosis of analog circuits and rotating machinery*. *Electronics* 11, 451 (2020).



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